

Tushar Kanti Dey

Full Stack Developer — DevOps Engineer

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Professional Summary

Pragmatic and product-minded Computer Science undergraduate who engineers software that endures beyond deadlines. From pixel-perfect frontends to battle-tested backend APIs, real-time systems, and infrastructure pipelines — I build with clarity, ship with confidence, and care deeply about what happens after deployment. Driven by a passion for system architecture, developer tooling, and crafting digital products that are fast, reliable, and genuinely useful at scale.

Core Competencies

Frontend Engineering	Next.js, React.js, TypeScript, Tailwind CSS, Component Architecture	Backend Engineering	Node.js, NestJS, Express.js, REST APIs, Authentication Systems
Data Systems	PostgreSQL, Prisma, Supabase, MongoDB, Redis, Schema Design	DevOps & Infrastructure	Docker, Kafka, CI/CD, Vercel, Render, Environment Management
Computer Science	Data Structures, Algorithms, OOP, DBMS, Operating Systems	Engineering Tooling	Git, GitHub, API Integration, Debugging, Performance Optimization

Experience

Jun 2024 – **Full Stack Engineer, AZMTH**

Present Impact: Engineered production-grade applications that raised the bar for reliability, performance, and team velocity — turning complex requirements into clean, deployable solutions that hold up under real-world load.

Key Contributions:

- Slashed frontend development time by **30%+** by architecting reusable component and service patterns that became team-wide engineering standards.
- Fortified deployment stability by **~40%** through battle-hardened CI/CD pipelines and proactive environment validation workflows.
- Designed scalable APIs and schema architectures capable of handling concurrent usage with surgical data flow efficiency.
- Championed production reliability through active runtime monitoring and rapid resolution of critical system failures before they became user-visible.

Technologies: Next.js, Tailwind CSS, Supabase, Git, Vercel

Feb 2025 – **Co-Tech Lead, CYCODERS**

Present Impact: Steered the engineering vision of a scalable CS club platform — building systems designed for longevity, team growth, and architectural clarity that survives changing requirements.

Key Contributions:

- Defined robust backend contracts and data models that trimmed future feature complexity by **~35%** — building once, building right.
- Owned full-stack feature delivery end-to-end: from schema inception to live deployment, with zero handoff friction.
- Uplifted **5+ contributors** through targeted mentorship that sharpened code quality and significantly cut review cycle overhead.
- Structured modular, intuitive codebases where onboarding is a feature, not a bottleneck.

Technologies: TypeScript, Node.js, Supabase, Git, Vercel

Mar 2025 – **Designer & Developer, Namespace**

Present Impact: Bridged the gap between design ambition and engineering reality — delivering UI systems that look intentional, feel cohesive, and perform relentlessly across every device class.

Key Contributions:

- Eliminated **40%+** of UI inconsistencies by building reusable component libraries backed by cohesive design tokens and a shared visual language.
- Transformed Figma blueprints into production-ready TypeScript implementations with near-zero rework cycles and high design fidelity.
- Pioneered structured implementation workflows that made design-to-engineering handoffs seamless, predictable, and repeatable.
- Tuned frontend performance across device classes — ensuring responsiveness from flagship phones to budget hardware without compromise.

Technologies: Figma, GIMP, Inkscape, TypeScript, Supabase, Docker, Google Cloud

Signature Projects

Aegis – Air-Gapped AIOps & Self-Healing Infrastructure

Overview: Crafted a local-first DevOps control plane that watches, thinks, and heals — autonomously detecting container failures, diagnosing root causes, and orchestrating recovery entirely without cloud dependency.

Engineering Contributions:

- Architected a headless NestJS control plane that captures Docker events and processes system telemetry with surgical precision.
- Engineered Kafka-based event pipelines to route crash logs, diagnostics, remediation commands, metrics, and audit trails in real time.
- Fused MongoDB, Redis, Docker Compose, and Python-powered AI/RL services into a fully offline automation backbone that requires no internet to function.
- Designed chaos simulation workflows to stress-test crash detection, recovery logic, and infrastructure resilience under adversarial failure conditions.

Tech Stack: NestJS, TypeScript, Kafka, Docker, MongoDB, Redis, Python, FastAPI.

DevMatch – Developer Matching & Collaboration Platform

Overview: Built a developer networking platform that goes beyond profiles — using GitHub activity, skill signals, and collaboration patterns to connect developers who are genuinely compatible for real projects.

Engineering Contributions:

- Wired GitHub OAuth authentication with automated developer profile enrichment for a frictionless, data-rich onboarding experience.
- Designed intelligent compatibility logic powered by language overlap, repository activity, project history, and contribution signals.
- Shipped real-time chat and match discovery flows that make developer collaboration feel immediate, natural, and purposeful.
- Deployed Redis caching and background job processing via BullMQ to keep expensive matching operations fast and non-blocking.

Tech Stack: Next.js, TypeScript, PostgreSQL, Prisma, NextAuth.js, Redis, BullMQ, Docker.

Synapse – Real-Time Software Intelligence Engine

Overview: Built a native desktop application that reads your codebase like a map — scanning local projects and rendering software architecture as a living, interactive graph that engineers can actually reason with.

Engineering Contributions:

- Developed a non-destructive project scanner that extracts files, frameworks, dependencies, Docker configurations, and architecture metadata with zero side effects on the source.
- Crafted an interactive graph interface with custom nodes, detail inspectors, minimap controls, and command-palette navigation for intuitive deep-dive exploration.
- Implemented smart framework and dependency detection across diverse stacks including Next.js, React, Vite, Tauri, Express, and NestJS.
- Engineered a project health scoring model that quantifies maintainability and surfaces actionable engineering quality signals at a glance.

Tech Stack: Tauri, Rust, React, TypeScript, Vite, Tailwind CSS, React Flow, Framer Motion.

WebScope Pro – Web Scraping & Data Processing Dashboard

Overview: Engineered a production-ready web scraping command center that extracts, stores, and surfaces structured website data with the discipline and reliability of an enterprise data pipeline.

Engineering Contributions:

- Built Puppeteer-driven scraping pipelines capable of fully rendering JavaScript-heavy SPAs before intelligent, structured extraction.
- Designed authenticated dashboards featuring scrape history, live statistics, paginated logs, and granular per-request result inspection.
- Modeled Prisma data schemas for users, request logs, and extracted content — optimized for query speed and long-term storage clarity.
- Hardened the application with validation layers, protected routes, real-time updates via Socket.IO, and reusable UI components built for maintainable scale.

Tech Stack: Next.js, TypeScript, Puppeteer, PostgreSQL, Prisma, NextAuth.js, Socket.IO, Zod.

CarePulse – Healthcare Management Platform

Overview: Delivered a healthcare operations platform where patient registration, appointment flows, admin workflows, secure document handling, and real-time SMS alerts work together in precise, clinical harmony.

Engineering Contributions:

- Built secure, intuitive patient registration and appointment management flows tailored for clinical accuracy and ease of use under high-stakes conditions.
- Developed a comprehensive admin panel enabling real-time booking approval, rescheduling, and cancellation with full operational audit visibility.
- Integrated Appwrite as a unified backend for database operations, authentication, and secure file storage workflows with minimal infrastructure overhead.
- Deployed Twilio-powered SMS notifications and Sentry monitoring to ensure consistent operational reliability across production environments.

Tech Stack: Next.js, TypeScript, Appwrite, Twilio, Sentry, Tailwind CSS, ShadCN UI.

Education

2023–Present **B.Tech in Computer Science Engineering**, *Adamas University*, Kolkata, India

Key Strengths

- Deep architectural intuition across full-stack systems — from API contract design to scalable data modeling, authentication, and infrastructure deployment.
- Proven ability to ship production-ready applications that withstand real-world load, with robust databases, deployment pipelines, and graceful failure handling.
- Genuinely versatile across frontend, backend, DevOps, automation, and system design — comfortable owning any layer of the stack and moving fluidly between them.
- Relentless debugging instinct paired with a performance-first mindset — writing code that is clean, observable, maintainable, and reliable by deliberate design.